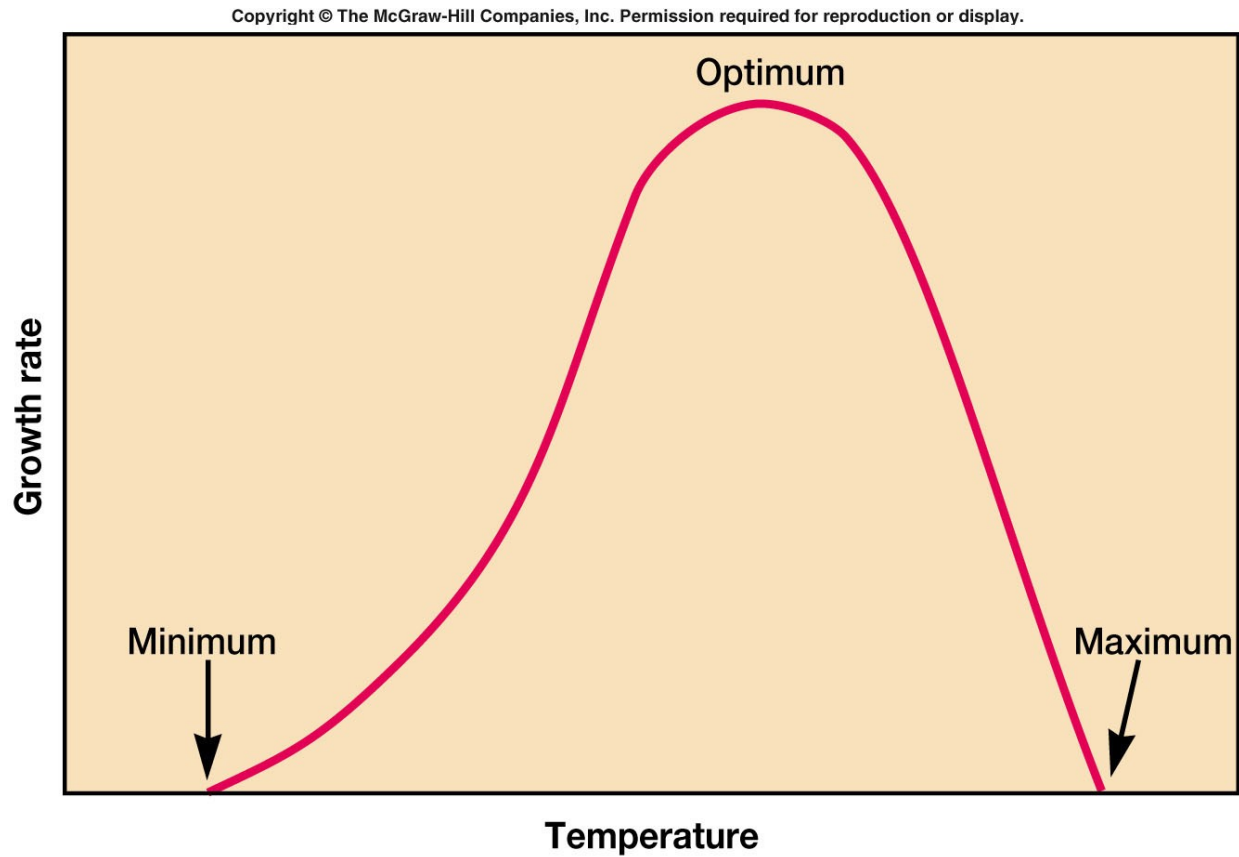


Microbial Growth

Microbial Growth - refers to the # of cells, not the size of the cells

- Requirements for Growth
 - Physical
 - Chemical

General response of microorganisms to temperature and pH.



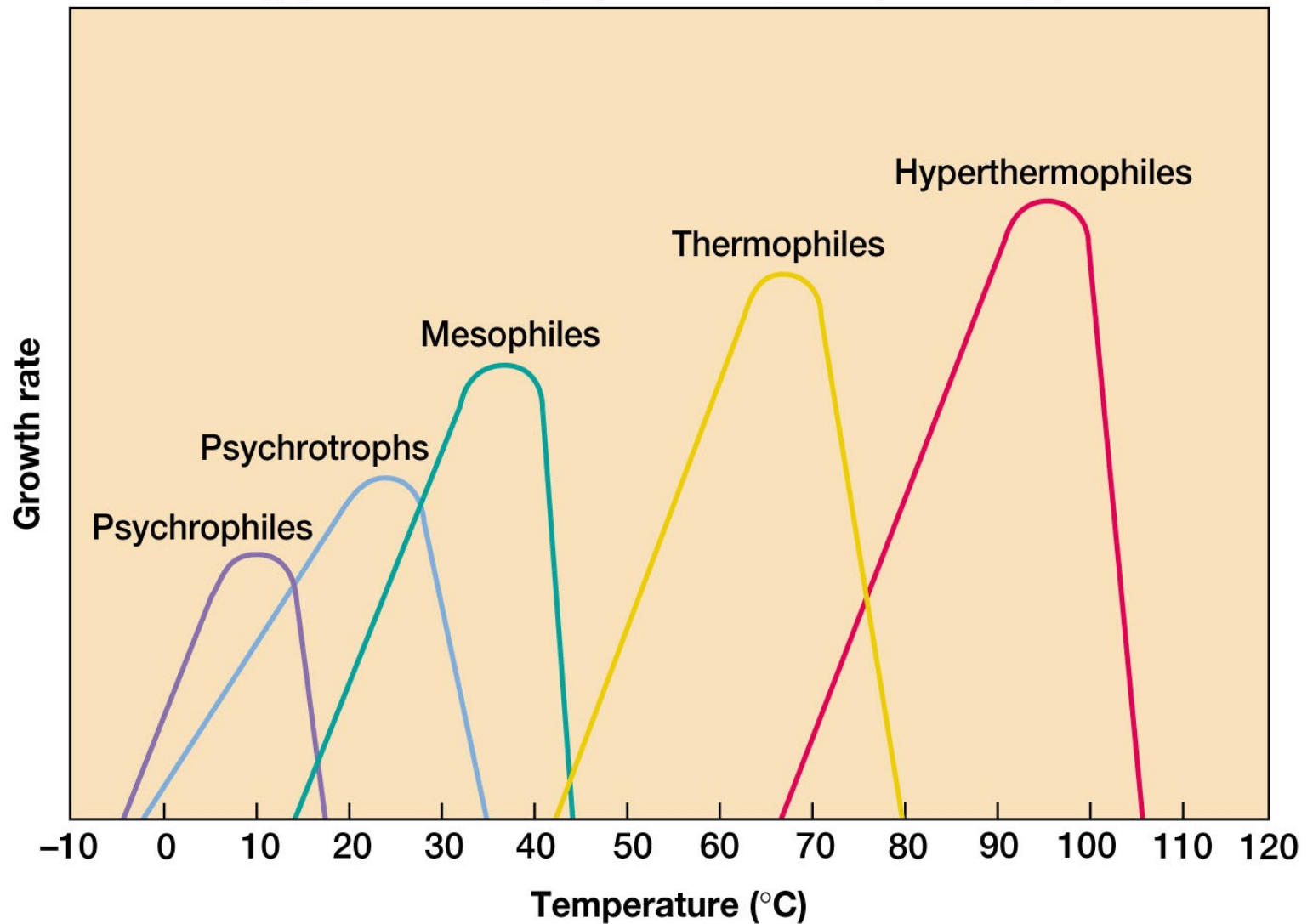
Depending on temperature preferences microorganisms are grouped into

Psychrophiles: Cold loving. Growth range 0-20°C with optimum at 10-15°C.

Mesophiles: Medium temperature loving. Range 20 to 40°C with optimum at 37°C.

Thermophiles: High temperature loving. Range 45-70°C with optimum at 60°C.

Extreme thermophiles: Growth range 80-120°C.



How thermophilic are thermophiles?

Growth at high temperature – what is high?

- Archaea grow to 121°C (opt about 113°C)
 - Many genera and species grow at or above 100°C
- Bacteria grow to 95°C (opt about 92°C)
 - Few species grow above 85°C
- Cyanobacteria grow to 68°C
- Fungi grow to 60-65°C
- Eucaryotic algae grow to 60°C
- Protozoa grow to 50°C

pH

- Most bacteria grow between pH 6.5 - pH 7.5
- Very few can grow at below pH 4.0
 - many foods, such as sauerkraut, pickles, and cheeses are preserved from spoilage by acids produced during fermentation

Soil pH

The pH or hydrogen ion concentration of natural environments varies from about 0.5 in the most acidic soils to about 10.5 in the most alkaline lakes.

Most free living prokaryotes can grow over a range of pH.

Depending on pH preferences microorganisms are classified into
acidophiles, neutrophiles and alkaliphiles.

Acidophiles - Optimum pH well below neutrality (7.0).

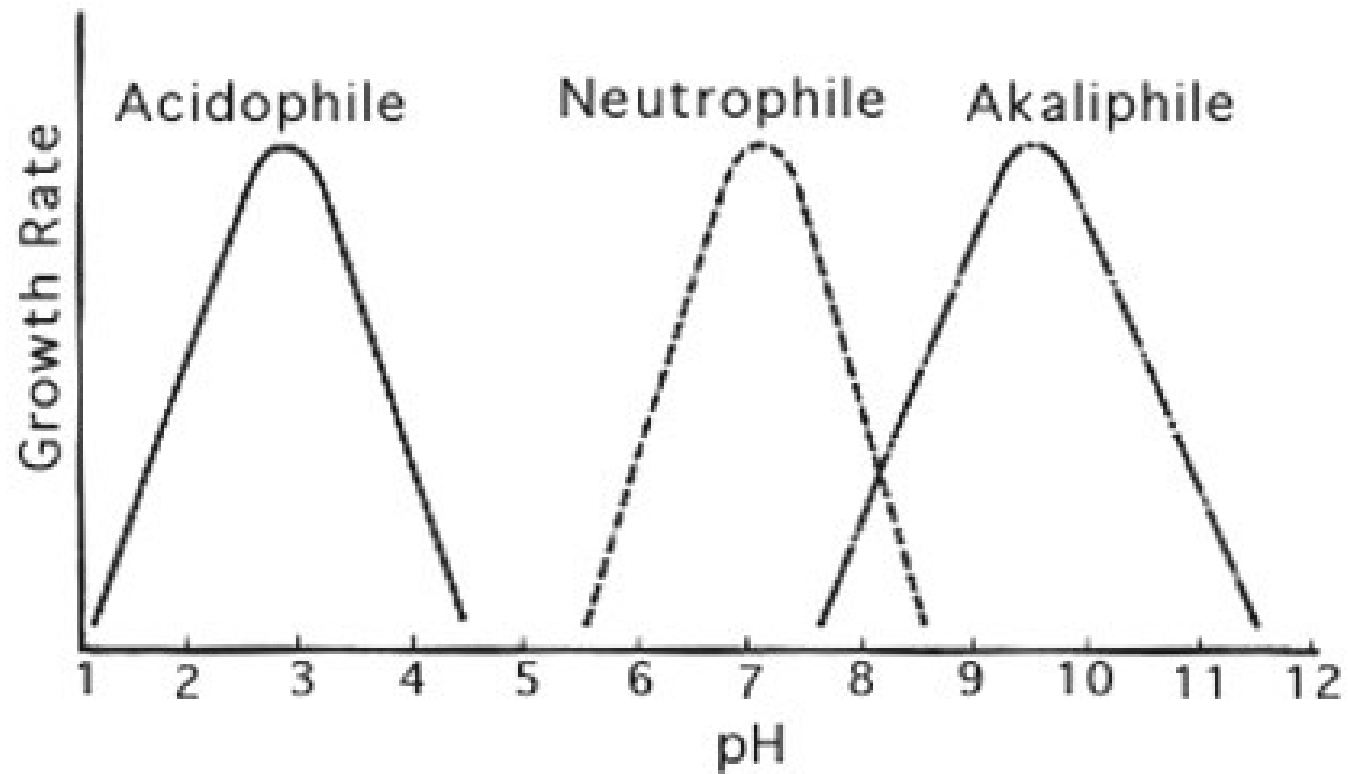
E.g. *Thiobaillus*, *Sulpholobus*, *Thermoplasma*

(obligate acidophiles). *Cyanidium* can grow at pH 0.

Neutrophiles - Grow best at neutral pH. Common genera such as *Bacillus*, *Pseudomonas* etc.

Alkaliphiles - Grow best under alkaline conditions.

Growth rate versus pH for three environmental classes of procaryotes.



Acidophiles

Archaea hold “world record”

- *Picrophilus* grows at pH -0.6 (and 60C)
- *Sulfolobus* grows at pH 1.0 (and up to 80C)

Bacteria

Thiobacillus grows at pH 1-3

Practically all extreme acidophiles are involved in sulfur or sulfide oxidation. They are therefore autotrophs (and aerobic) and produce sulfate (in the form of sulfuric acid) as an end product of metabolism. They are therefore resistant to low pH only due to the sulfate ion and they will not grow at low pH induced with any other acid.

Alkaliphiles

Obviously grow at high pH – how high?

No natural environments above pH 13 so this represents the upper limit.

Osmotic Pressure

- Microbes obtain almost all their nutrients in solution from surrounding water
 - isotonic
 - hypertonic
 - hypotonic

SOIL MOISTURE CONTENT

Availability of water (a_w - known as water activity) is a critical factor that affects the growth of all cells.

The water activity of pure water is 1

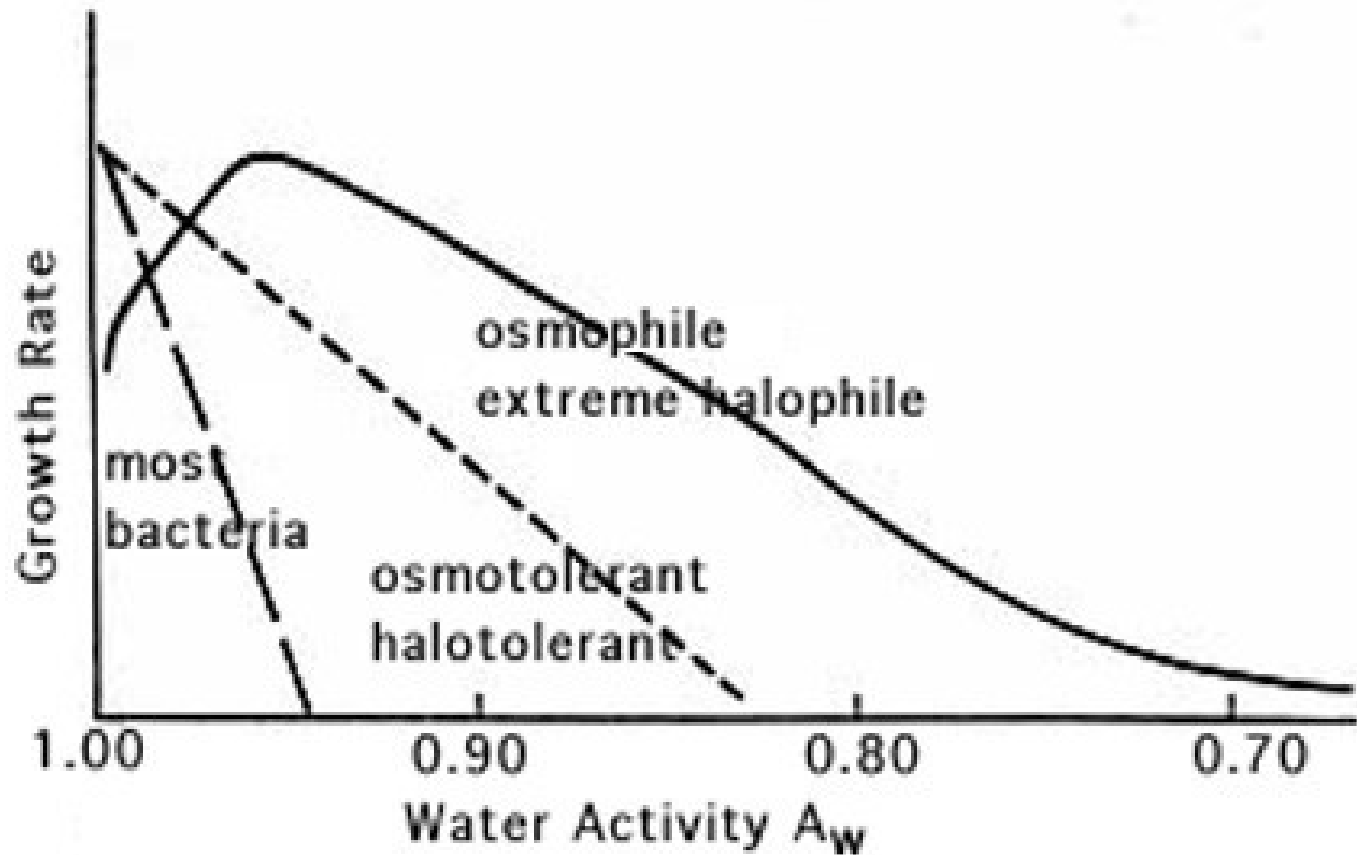
Microorganisms live over a range of a_w from 1 to 0.7.

Water activities of agricultural soils range between 0.9 to 1.

Microorganisms live in dry and arid environments are known as **xerophiles**.

A_w of blood is 0.99, sea water = 0.98,
maple syrup = 0.90, great salt lake =
0.75.

Limiting a_w for growth of certain procaryotes.



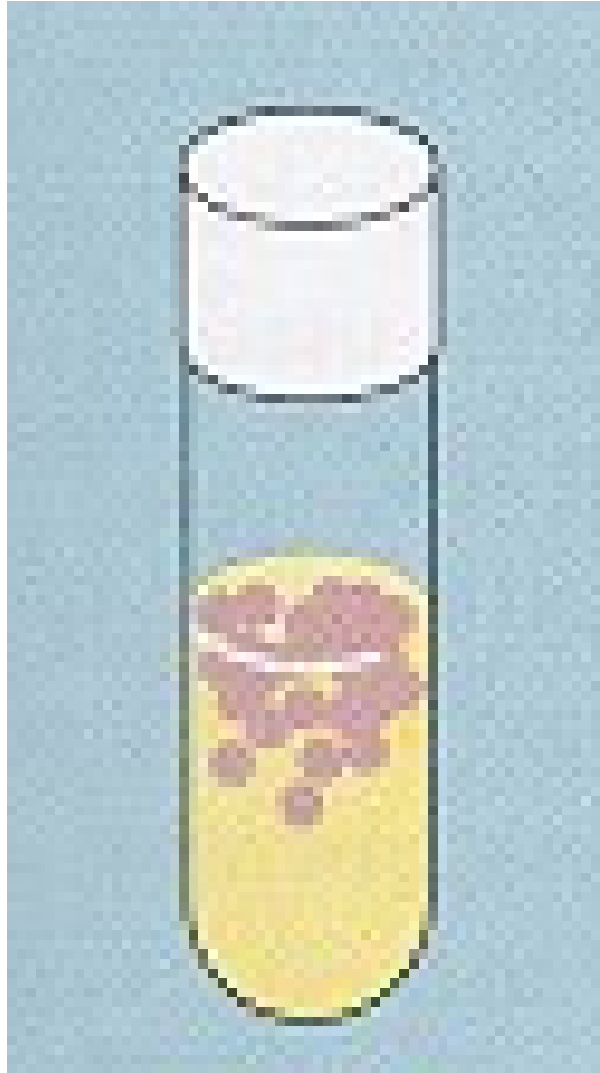
Chemical Requirements

- Macro & Micro Elements
- C, H, O, P, K, N, S Ca, Fe Mg Na, Cl

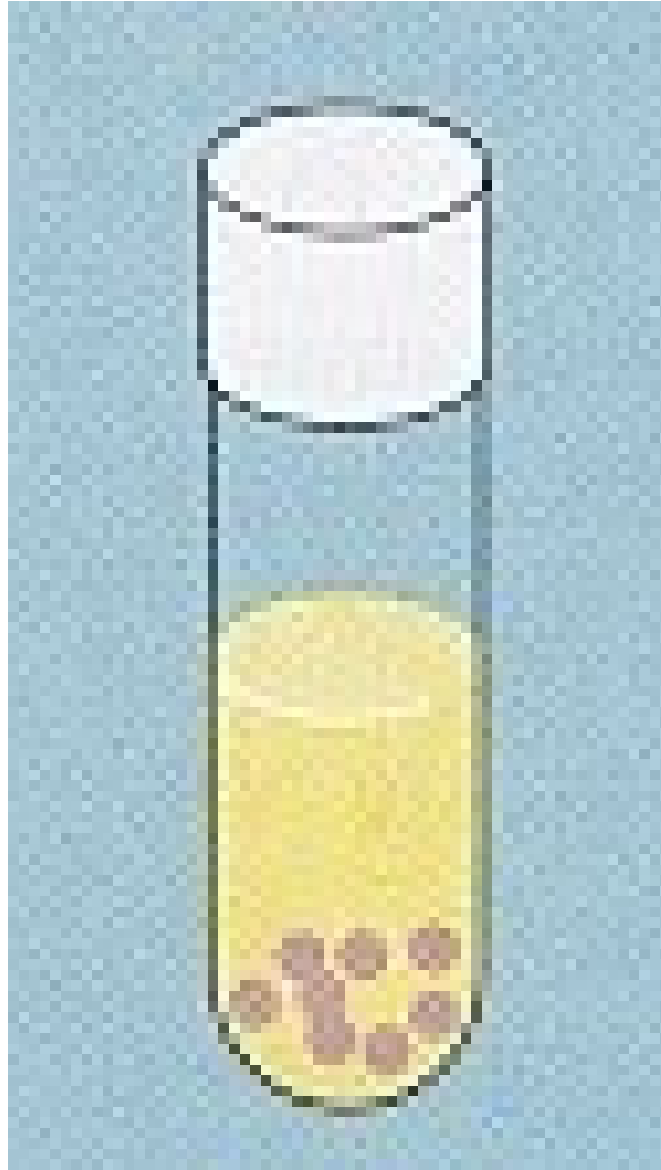
Oxygen

- Bacteria can be classified base on their oxygen requirements

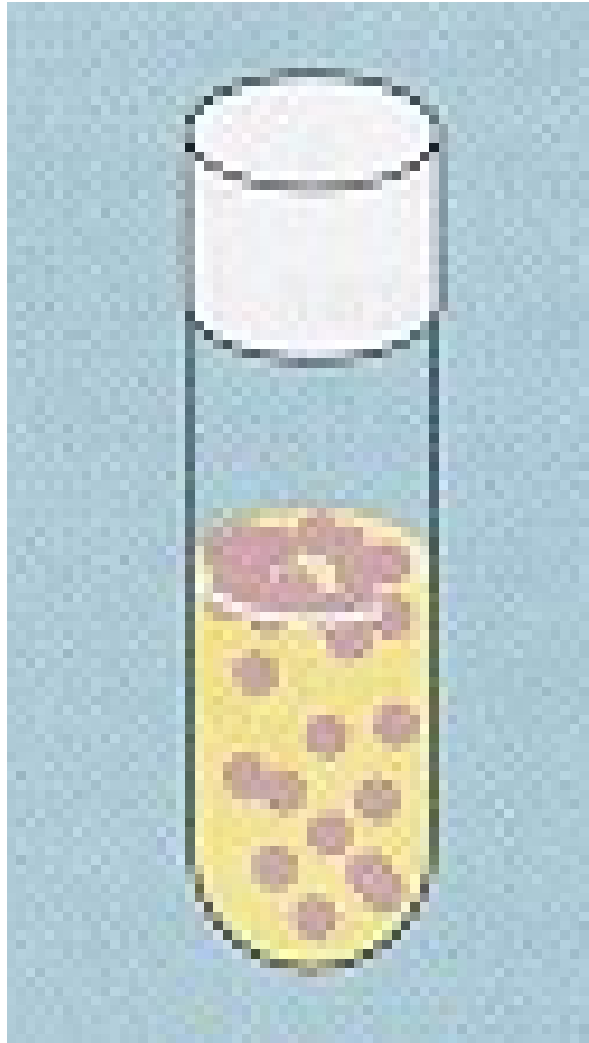
1. Obligate Aerobes



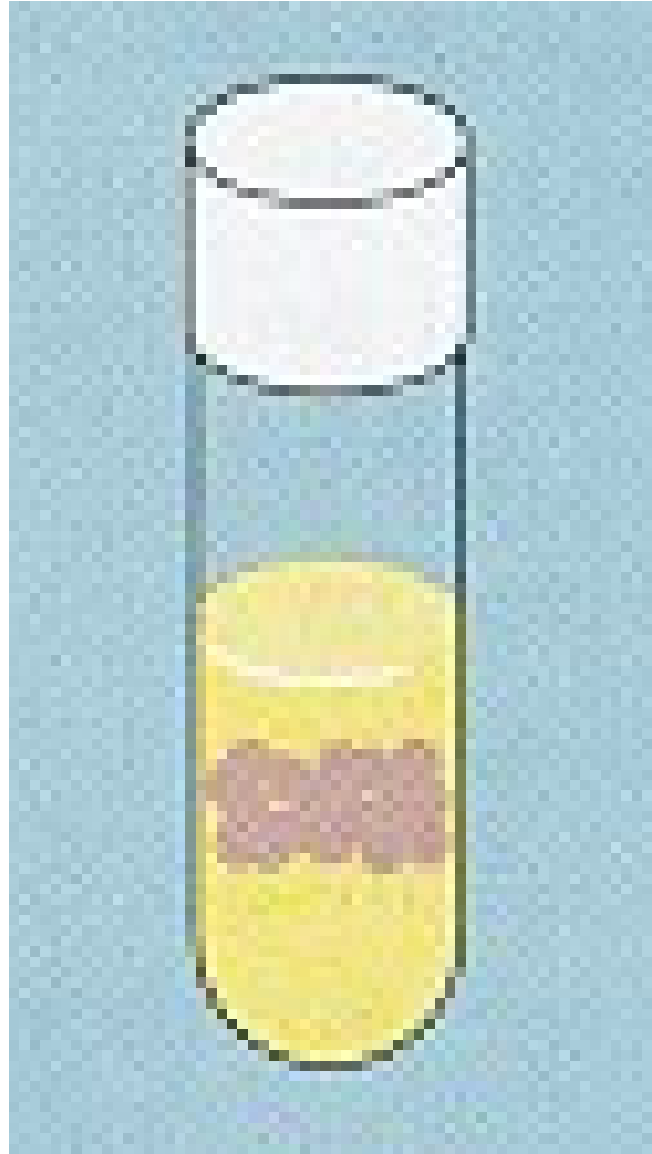
2. Obligate Anaerobes



3. Facultative Anaerobes



4. Microaerophilic



Oxygen is lethal to some organisms

- All organisms produce superoxide (O_2^-)
- Superoxide is toxic to cells (steals electrons)
- Superoxide must be neutralized

Superoxide dismutase

- $\text{O}_2^- + \text{O}_2^- + 2 \text{H}^+ \longrightarrow \text{H}_2\text{O}_2 + \text{O}_2$
- Hydrogen peroxide is also toxic to cells and it must be neutralized

Catalase

- $2 \text{H}_2\text{O}_2 \text{ -----} > 2 \text{H}_2\text{O} + \text{O}_2$
- Obligate Anaerobes lack:
 - Superoxide dismutase (SOD)
 - Catalase

Culture Media

- 1. Chemically Defined
 - the exact chemical composition is known
 - used to grow fastidious organisms
- 2. Complex Media
 - exact chemical composition is not known
 - most bacteria and fungi are grown with this

Special Culture Techniques

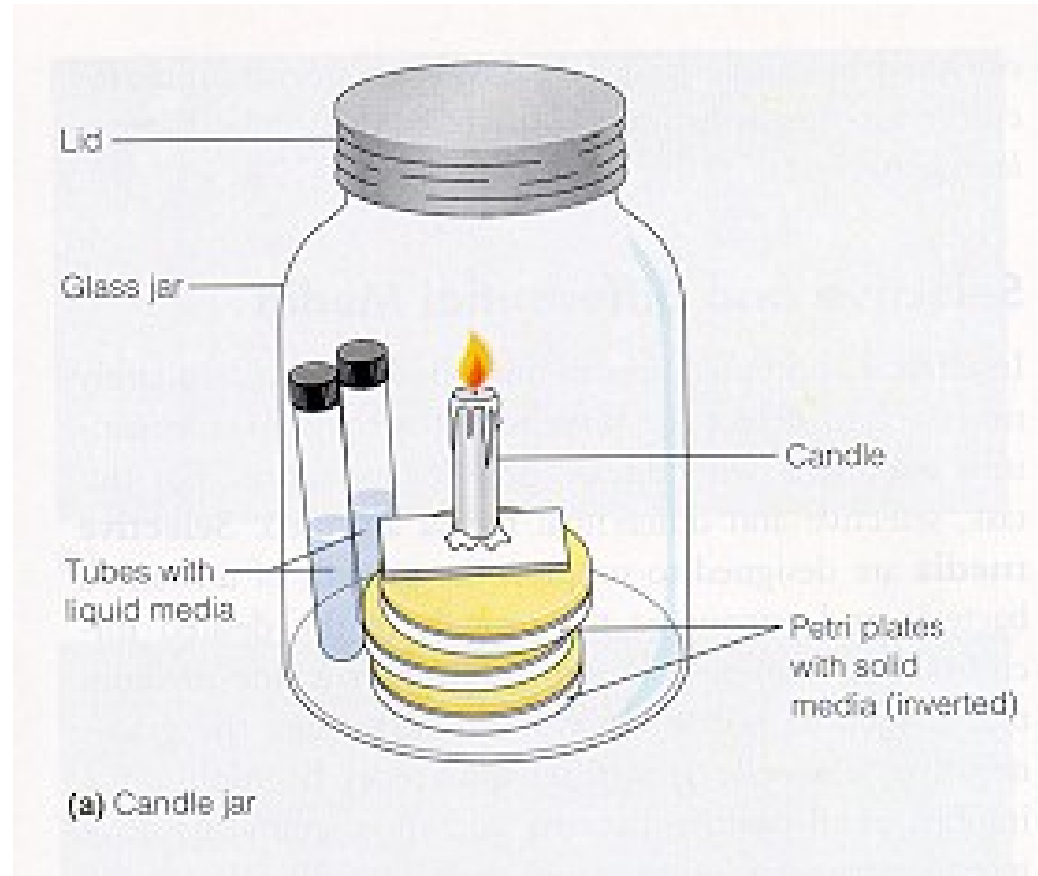
- 1. Anaerobic Bacteria
 - a. Reducing Media
 - b. Anaerobic Container
 - c. Agar Stab

Special Culture Techniques

- 2. Microaerophilic Bacteria
 - grow best under reduced O_2 levels and increased CO_2 levels
 - Normal Atmosphere 21 % O_2
 - .3 to .03 % CO_2

Microaerophilic Bacteria

A. Candle Jar

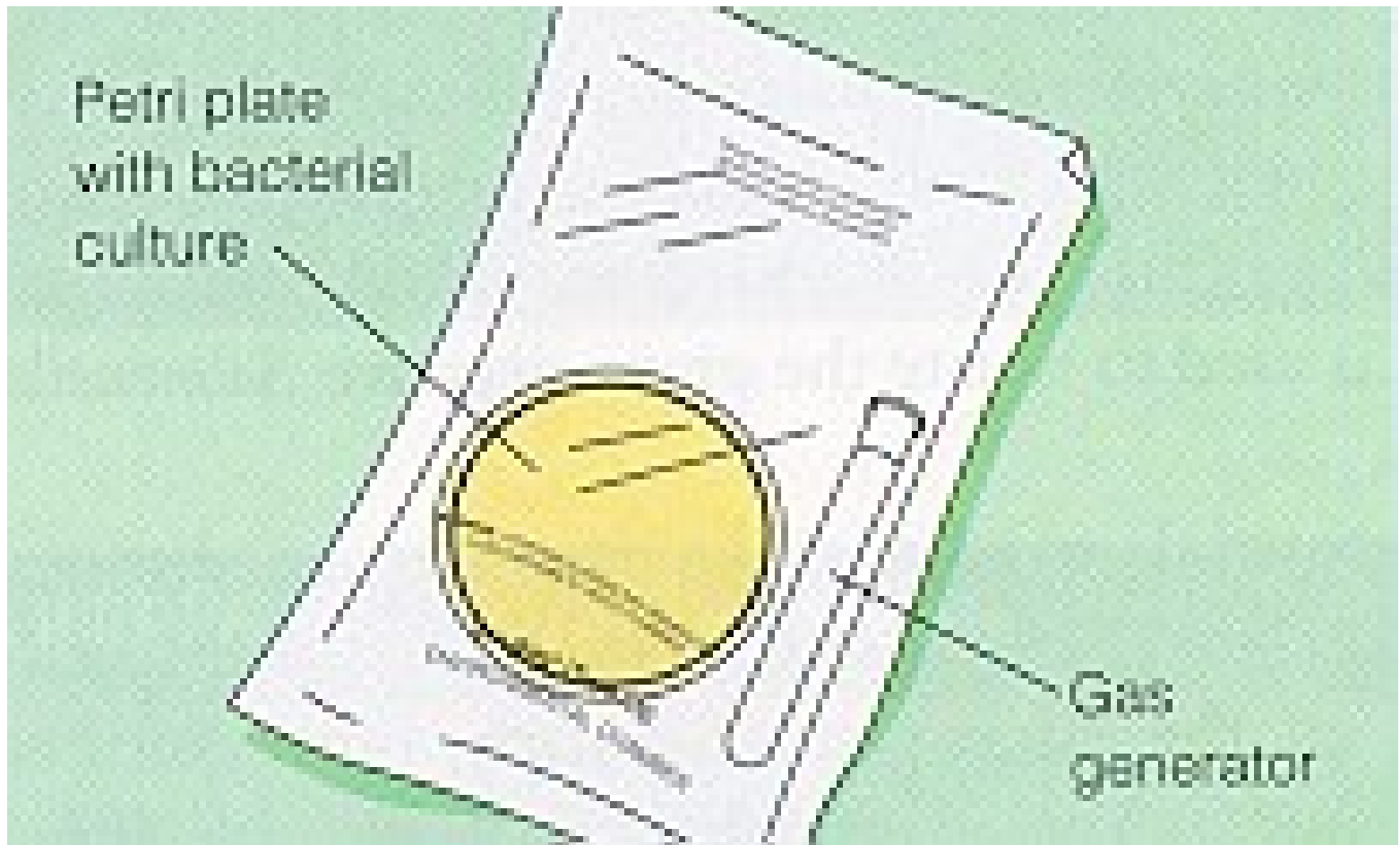


16 % O_2

4 % CO_2

Microaerophilic Bacteria

B. CO₂ Generating Packet

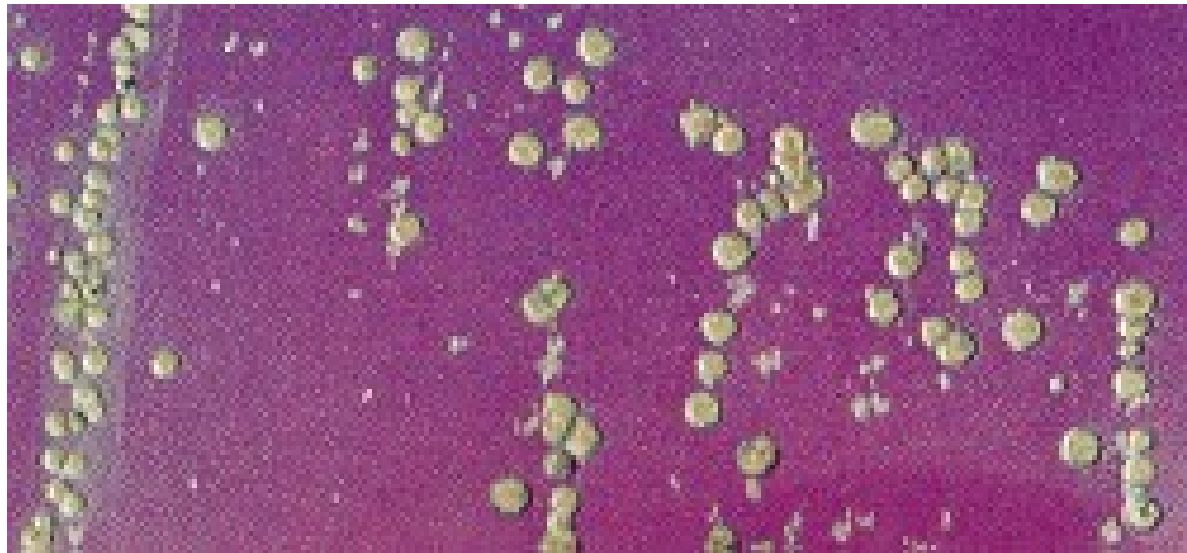


Selective Media

- Inhibits the growth of some bacteria while selecting for the growth of others
- Example:
 - Brilliant Green Agar
 - dyes inhibit the growth of Gram (+) bacteria
 - selects for Gram (-) bacteria
 - Most G.I. Tract infections are caused by Gram (-) bacteria

Selective Media

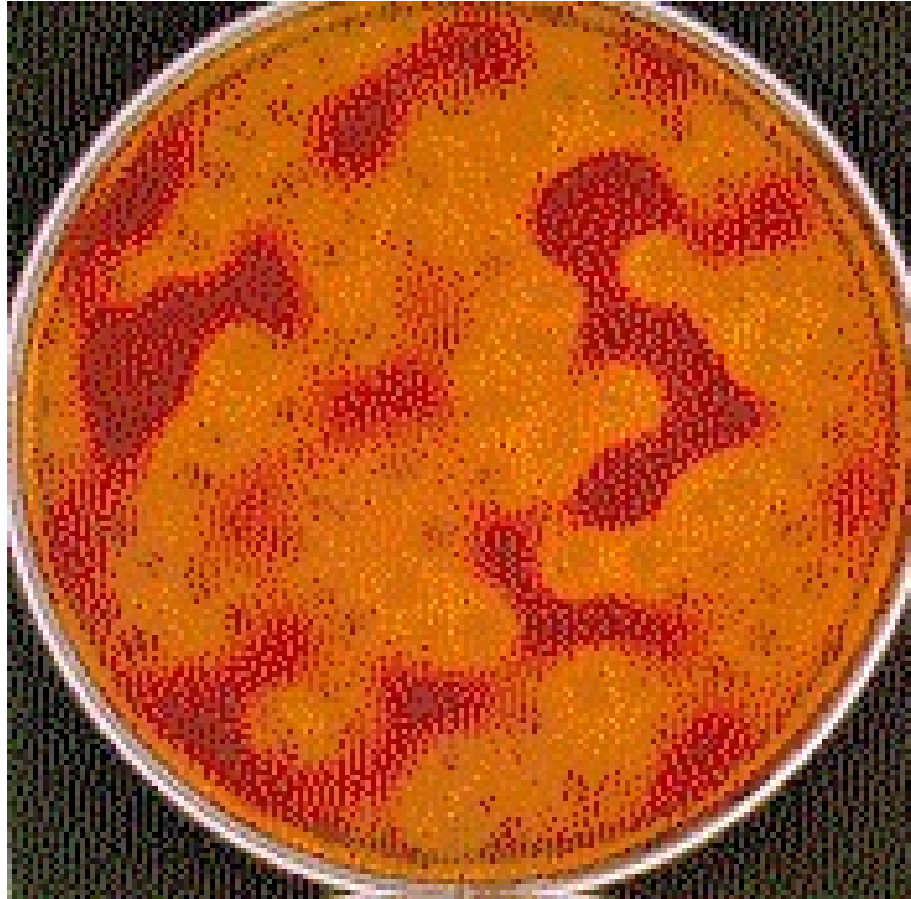
- EMB (Eosin Methylene Blue)
 - dyes inhibit Gram (+) bacteria
 - selects for Gram (-) bacteria
 - G.I. Tract infections caused by Gram (-) bacteria



Differential Media

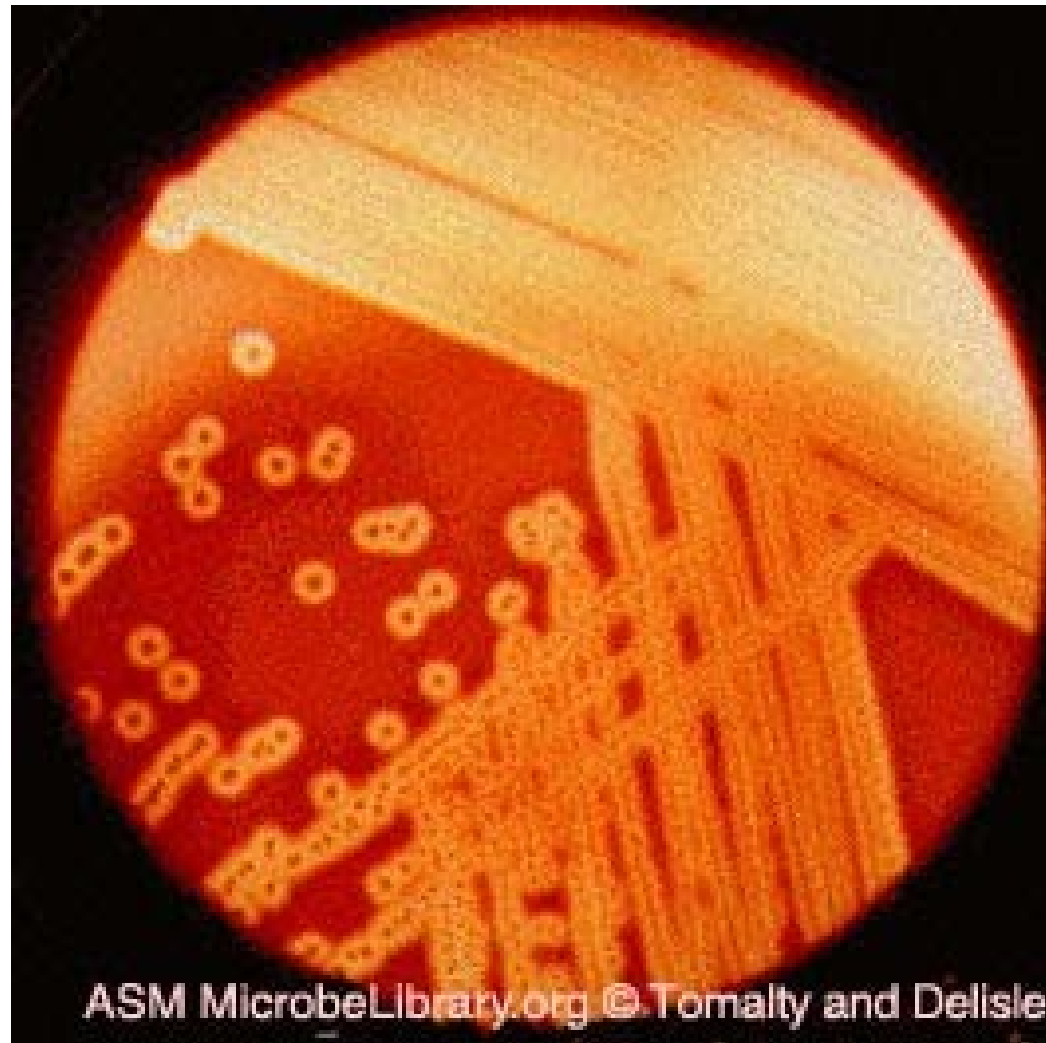
- Differentiates between different organisms growing on the same plate
- Example:
 - Blood Agar Plates (TSA with 5% sheep blood)
 - used to differentiate different types of *Streptococci*

Alpha Hemolytic *Streptococci*



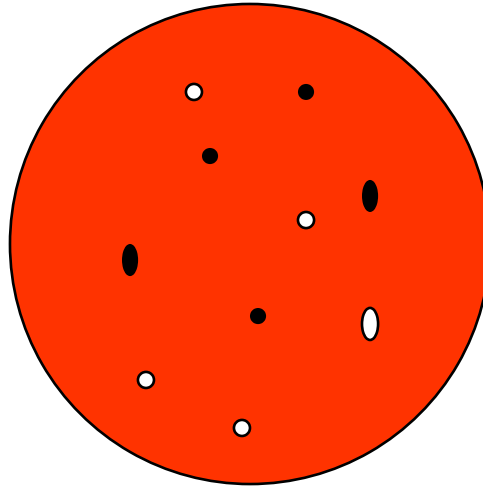
Incomplete lysis of RBC's

Beta Hemolytic *Streptococci*



Complete lysis of RBC's

Gamma Hemolytic *Streptococci*



No lysis of RBC's

Selective and Differential Media

- Mannitol Salt Agar
 - used to identify *Staphylococcus aureus*
- Mannitol Salt Agar
 - High salt conc. (7.5%) inhibits most bacteria
 - sugar Mannitol
 - pH Indicator (Turns Yellow when acid)

Selective and Differential Media

- MacConkey's Agar
 - used to identify *Salmonella*
- MacConkey's Agar
 - Bile salts and crystal violet (inhibits Gram (+) bacteria)
 - lactose
 - pH Indicator

Many Gram (-) enteric non-pathogenic bacteria can ferment lactose, *Salmonella* can not